



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

DATASET OVERVIEW

The Foundation

3,900

Total Purchases

Transactions analyzed

18

Data Points

Features per customer

50

Locations

Geographic spread

25

Products

Items tracked



Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Exploration

Analyzed structure and summary statistics

03

Missing Data

Imputed 37 missing review ratings with median

04

Feature Engineering

Created age groups and purchase frequency metrics

05

Database Integration

Loaded cleaned data into PostgreSQL

Key Business Insights

Revenue by Gender

Male customers: **\$157,890**

Female customers: **\$75,191**

Males generate 2.1x more revenue

Shipping Preferences

Express: **\$60.48** avg

Standard: **\$58.46** avg

Express users spend slightly more



Top-Rated Products



Gloves

Highest average rating: 3.86



Sandals

Strong performance: 3.84



Boots

Customer favorite: 3.82



Hat

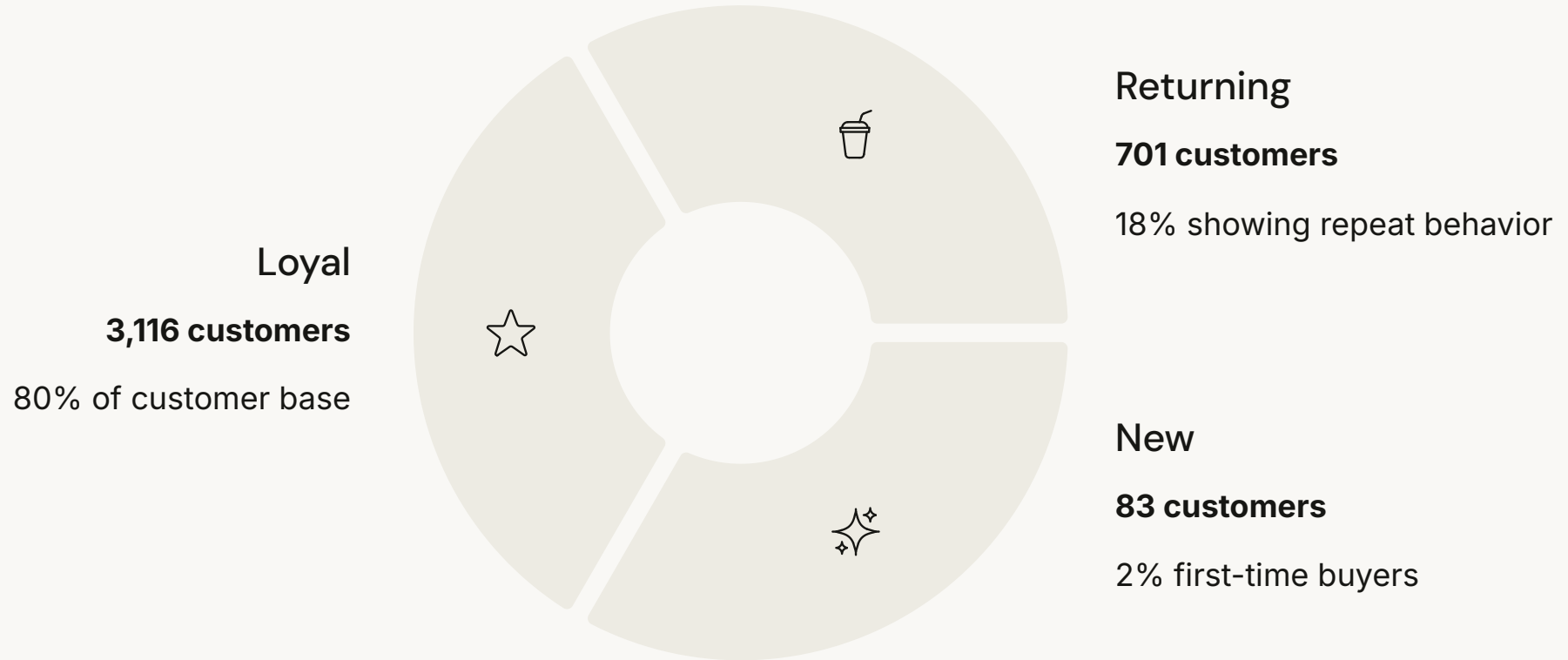
Solid rating: 3.80



Skirt

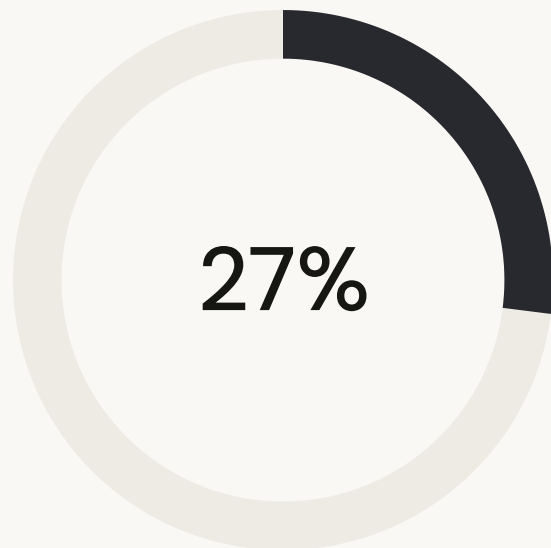
Well-received: 3.78

Customer Segmentation



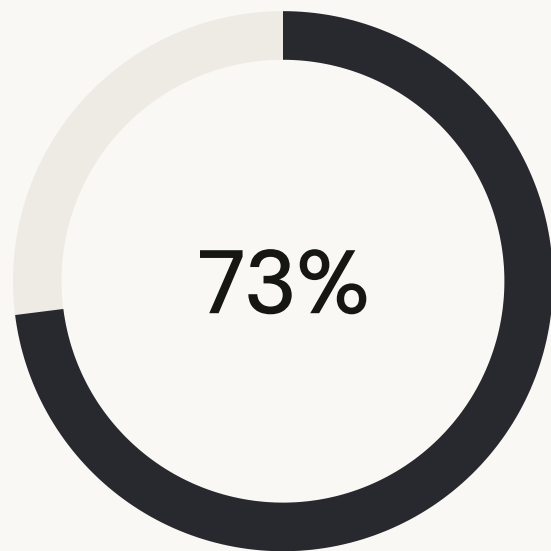
Strong loyalty foundation with opportunity to convert returning customers

Subscriber Impact



Subscribers

1,053 customers

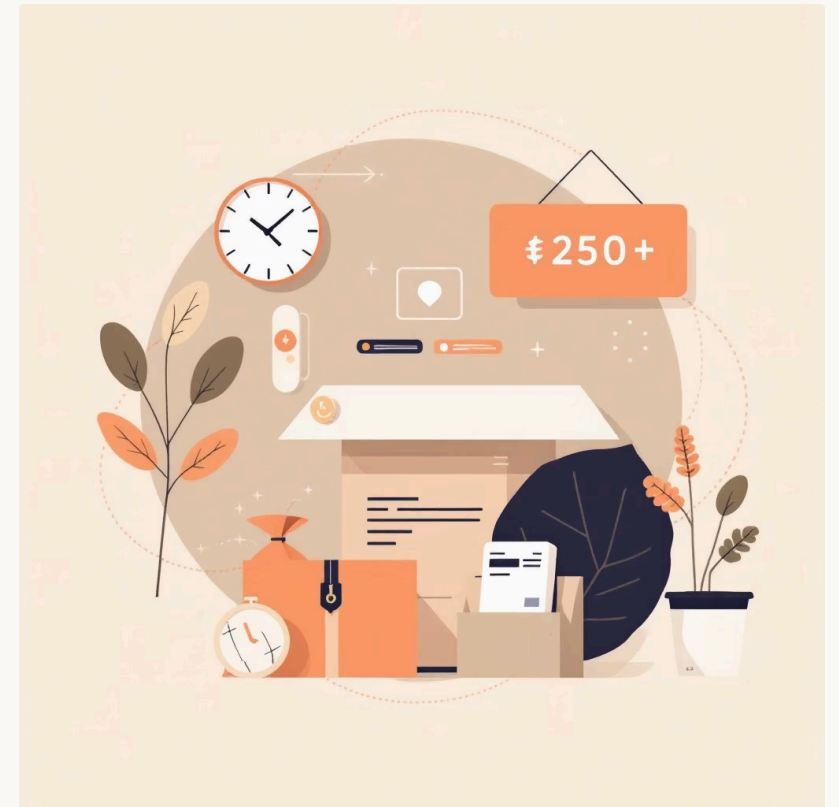


Non-Subscribers

2,847 customers

Average spend nearly identical: **\$59.49** vs **\$59.87**

Repeat buyers (>5 purchases): **958** subscribers vs **2,518** non-subscribers





Revenue Distribution

Young Adult

\$62,143

Highest revenue contributor

Middle-aged

\$59,197

Strong purchasing power

Adult

\$55,978

Consistent spending

Senior

\$55,763

Stable segment

Young adults drive the highest revenue, but all segments show balanced contribution

Interactive Dashboard

Key Metrics

3.9K customers

\$59.76 avg purchase

3.75 avg rating

Category Performance

Clothing leads revenue

Accessories strong second

Balanced distribution

Interactive Filters

Subscription status

Gender & category

Shipping type

Strategic Actions

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base

Loyalty Programs

Reward repeat buyers to strengthen the 80% loyal customer segment

Review Discount Policy

Balance sales boosts with margin control—839 high spenders used discounts

Product Positioning

Highlight top-rated items (Gloves, Sandals, Boots) in marketing campaigns

Targeted Marketing

Focus on young adults and express-shipping users for maximum ROI